Nurture: A Leakage-Free Machine Learning and Rule-Based Decision Support System for Antenatal Care

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***Abstract*—Antenatal records are spread across visits, laboratory reports, prescriptions and diet assessments, so the people who use them rarely see one coherent picture. We present Nurture, a decision support system built on six linked synthetic datasets describing 200 Indian pregnancies, 2,048 antenatal visits and 12,863 laboratory tests. An audit of the data pipeline found seven defects, including an outcome generator that labelled every pregnancy as gestational diabetes and a delivery rule that inflated preterm births to 31%; after correction the cohort shows 17.5% preterm birth and 27.5% gestational diabetes (GDM). We train visit-level models that use only information available up to each visit, validated with 5-fold cross-validation grouped by pregnancy. The models reach ROC-AUC 0.682 for preterm birth, 0.706 for NICU admission and 0.765 for GDM before the 24-week glucose test, within 0.04 of the best score achievable on this data. Prescribing safety, alerts and test schedules are handled by a deterministic rule engine, and every model output is confirmed or overridden by a doctor with an audit trail. A role-based web application for patients, doctors and dieticians responds in 10–130 ms per screen.**

***Keywords*—maternal health, antenatal care, preterm birth, gestational diabetes, clinical decision support, data leakage, grouped cross-validation**

I. Introduction

An estimated 13.4 million babies were born preterm in 2020, and India recorded the largest number of any country, about 3.02 million [1]. India's maternal mortality ratio fell to 97 deaths per 100,000 live births in 2018–20 [2], but antenatal care still depends on clinicians piecing together a woman's history at each visit. The WHO model of antenatal care recommends at least eight contacts per pregnancy [3]; each contact adds blood pressure readings, laboratory results, prescriptions and dietary advice that are usually stored and viewed separately.

Most machine learning work on maternal risk uses a single flat table of vital signs [4]. Real antenatal data is longitudinal: it grows visit by visit, and a model that silently uses later visits or the delivery record will look accurate in testing and fail in practice. Prediction is also only one part of the problem—a doctor must know which medicine conflicts with an allergy, which tests are overdue and what the patient should eat.

This work makes five contributions:

- An audit of a six-dataset synthetic antenatal cohort that found and fixed seven generator and pipeline defects.
- Leakage-free, visit-level features in which every value is computable at the visit it describes.
- Three risk models (preterm birth, NICU admission, GDM) with grouped validation and a ceiling analysis that measures how close each model is to the best achievable score.
- A deterministic rule engine for prescribing safety, alerts, test schedules and visit intervals, combined with doctor review of every model output.
- A fast role-based web application for patients, doctors and dieticians, including appointment booking.

II. Related Work

Ahmed *et al.* [4] collected blood pressure, blood glucose, body temperature and heart rate from pregnant women in rural Bangladesh to predict a three-level maternal risk class. The resulting dataset is widely used, but each row is a single reading with six features, so it cannot represent change across visits.

Sharifi-Heris *et al.* [5] reviewed 13 studies that predicted preterm birth from health records. Reported ROC-AUC ranged from 0.62 to 0.98, with most between 0.65 and 0.85; complex models often did not outperform logistic regression, and few studies justified their metrics, handled class imbalance or validated externally.

Artzi *et al.* [6] predicted GDM from nationwide electronic health records at the start of pregnancy with ROC-AUC 0.85, and a nine-question model still reached 0.80, showing that a small number of history variables such as previous GDM carry most of the signal. None of these systems combines prediction with prescribing safety checks and role-specific views, which is the gap this project addresses.

III. Dataset and Data Quality

A. Six linked datasets

The cohort follows 200 pregnancies from booking to delivery across six datasets linked by Pregnancy_ID, with event tables also linked to the exact visit through Visit_ID

(Table I). Reference values follow Indian guidance: anaemia management from Anemia Mukt Bharat [7], foods and diet patterns from ICMR-NIN [8] and medicines from the National List of Essential Medicines [9]. Visits run from March 2025 to June 2027 and maternal age from 19 to 40.

TABLE I. STRUCTURE OF THE SIX LINKED DATASETS

Dataset	*One row is*	*Rows × cols*
D1 Patient & pregnancy	one pregnancy	200 × 48
D2 ANC visits	one visit	2,048 × 39
D3 Laboratory	one test (29 types)	12,863 × 19
D4 Medication	one medicine event	874 × 34
D5 Nutrition	one assessment	2,048 × 36
D6 Outcomes	one delivery	200 × 22

Reference tables accompany the event data: a catalogue of 32 medicines with an allergy crosswalk and 13 interaction rules, and 60 Indian foods with 18 nutrition rules and 12,123 generated food recommendations.

B. Defects found and fixed

Inspecting the generators and the pipeline output revealed seven defects:

- **Preterm labels.** D2 chose a delivery week but never saved it, so D6 used the week of the last antenatal visit; a birth at 38 weeks whose last check-up was at 36 weeks became preterm. D2 now writes the planned week.
- **GDM labels.** D6 marked GDM whenever a screening test existed. Screening is universal, so all 200 pregnancies had GDM. GDM now requires an abnormal result.
- **Gestational days** at delivery were always zero; they are now drawn from a separate random stream.
- **Diet pattern drift.** The keyword “omnivore” did not match “Omnivorous”, so diet type was re-randomised at each visit.
- **Lab values.** Preprocessing label-encoded the mixed text/number Result column, destroying haemoglobin and glucose values; a numeric copy scaled per test is now kept.
- **Date parsing.** Text columns such as Delivery_Mode were parsed as dates; only true date columns are parsed now.
- **Missing values.** The literal value “None” was read as missing; only empty cells are now treated as missing.

All generators were re-run with the original random seed. The D2 visit table is byte-identical to the original, so D3–D5 remain valid, while the outcome rates change as shown in Fig. 1.

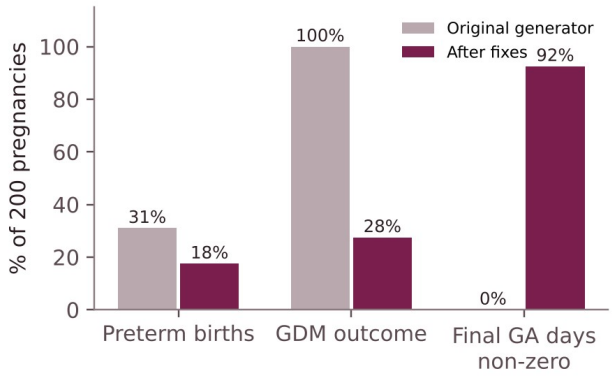


Fig. 1. Outcome rates before and after correcting the data generators.

IV. METHODOLOGY

Fig. 2 shows the system. The six datasets feed two parallel paths—a machine learning pipeline that produces risk predictions and a rule engine that produces safety checks and care plans—and both are served to three role-based views, with the doctor’s confirmed risk passed on to the patient and the dietician.

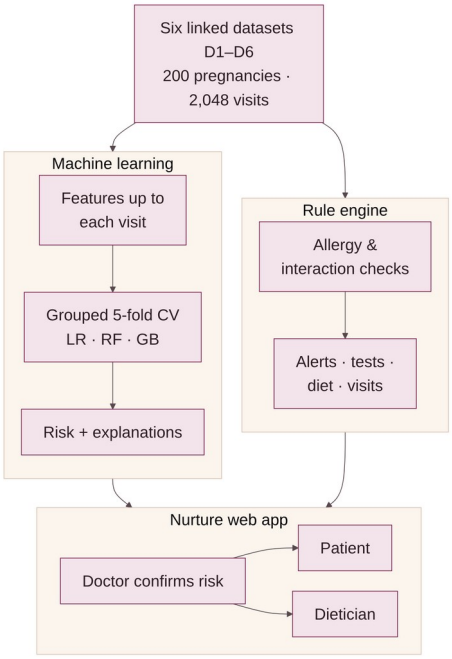


Fig. 2. System architecture: datasets feed a leakage-free ML pipeline and a rule engine, both served to three role-based views.

A. Prediction targets

Three binary outcomes are predicted: preterm birth before 37 weeks (35 of 200 pregnancies), NICU admission of the newborn (14 of 200) and GDM (55 of 200). GDM is diagnosed by the 24–28-week glucose test, so its model is trained only on visits before 24 weeks; the clinical question is which women to watch closely before the test. The preterm model uses visits before 37 weeks, because a pregnancy that has reached 37 weeks cannot be preterm.

B. Leakage-free visit-level features

Each of the 2,048 visits becomes one training row with 49 features in five groups: baseline history from D1 (19, e.g. previous pre-eclampsia, multiple pregnancy, BMI), vital signs at the visit (7), cumulative visit history such as the highest systolic pressure and the number of high-risk visits so far (9), laboratory results up to the visit such as lowest and latest

haemoglobin and proteinuria (9), and medicines started so far (5). Cumulative values are computed with running maxima, sums and forward fills within each pregnancy, so no value uses a later visit or the delivery record.

C. Models and validation

Logistic regression with class weighting, a random forest [10] of 400 trees and histogram gradient boosting were compared using scikit-learn [11]. Validation is 5-fold StratifiedGroupKFold with the pregnancy as the group, so a woman's visits never appear in both training and test folds. The model with the highest out-of-fold ROC-AUC is selected, and every patient is scored by the fold model that did not see her. Accuracy-type metrics are reported for one prediction per pregnancy, taken at the last visit up to 28 weeks (24 weeks for GDM), at the threshold maximising sensitivity plus specificity.

Risk bands are set on out-of-fold predictions: Moderate above the 75% specificity point and High above the 90% specificity point. Each prediction is explained by occlusion—every feature is reset to the cohort median and the change in predicted risk is shown to the doctor.

Because the generator's rules are known, the best achievable ROC-AUC can be computed by scoring each pregnancy with the true probability the generator used. For preterm birth this probability depends only on a baseline risk score and multiple pregnancy; for GDM, only on previous GDM and pre-existing diabetes.

D. Rule engine and doctor review

Safety-critical logic is deterministic rather than learned, because an allergy conflict must be caught every time and be traceable to a source table. The engine checks each medicine against the patient's allergy class, the 13 interaction rules, pregnancy-avoid flags and renal cautions; raises alerts for raised or severe-range blood pressure, proteinuria, anaemia (Hb below 11 g/dL), abnormal glucose, low SpO2, reduced fetal movement and malpresentation after 36 weeks; lists overdue and upcoming tests by gestational week; and compares weight gain with the IOM range for the woman's BMI [12]. Rule-based medication suggestions, such as low-dose aspirin for pre-eclampsia risk factors [13], are shown for the doctor to accept or ignore.

A visit is flagged for risk review when it has abnormal results, a serious alert, or a model band that disagrees with the visit's recorded risk grade. The doctor confirms or changes each band; a written reason is required for any change, and every decision is stored with the model's original band. The confirmed band, not the raw model output, drives the patient's message, the dietician view and the recommended visit interval.

E. Appointments and implementation

The recommended visit interval is four weeks before 28 weeks, two weeks from 28 to 36 and weekly after 36, halved for high risk. Patients book 20-minute slots; a slot cannot be double-booked and a patient can hold one booking per day. Ticking a danger sign such as bleeding or reduced fetal movement directs the patient to hospital and to the 108 and 102 ambulance services.

The back end is a FastAPI service in Python that loads the data and models once, computes all 2,048 risk scores in one vectorised pass and pre-computes explanations in a background thread. The front end is plain HTML, CSS and JavaScript with hand-drawn SVG charts; each screen needs one gzip-compressed API call.

V. Results

A. Model performance

Table II summarises the selected models. The random forest was best for preterm birth and NICU admission; logistic regression was best for GDM, consistent with the observation in [5] that simple models are often competitive. All three models rank patients well above chance (Fig. 3).

TABLE II. Out-of-Fold Performance of the Selected Models

Metric	*Preterm*	*NICU*	*GDM*
Model	RF	RF	LR
ROC-AUC (visits)	0.682	0.706	0.765
Best possible	0.696	—	0.801
Accuracy	0.63	0.72	0.81
Always-no accuracy	0.83	0.93	0.73
Balanced accuracy	0.65	0.72	0.74
Sensitivity	0.69	0.71	0.60
Specificity	0.62	0.72	0.88
F1	0.39	0.26	0.63

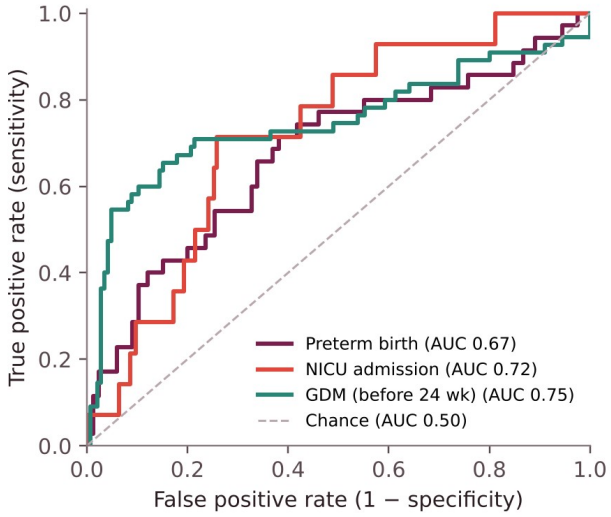


Fig. 3. ROC curves, one prediction per pregnancy (last visit up to 28 weeks; up to 24 weeks for GDM).

Accuracy for preterm birth and NICU admission is below the always-no baseline. This is expected for rare outcomes: a model that never predicts preterm birth is 83% accurate and detects no cases, whereas the selected threshold detects 69% of preterm births. Balanced accuracy and ROC-AUC are therefore the appropriate measures.

B. Ceiling analysis

Scoring each pregnancy with the generator's true probabilities gives ROC-AUC 0.696 for preterm birth and 0.801 for GDM (Fig. 4). Our models reach 0.682 and 0.765, within 0.04 of the limit. In the generator, 16 of the 35 preterm births occur in women with no risk factor, so no model can identify them; a substantially higher score on this data would indicate leakage rather than skill.

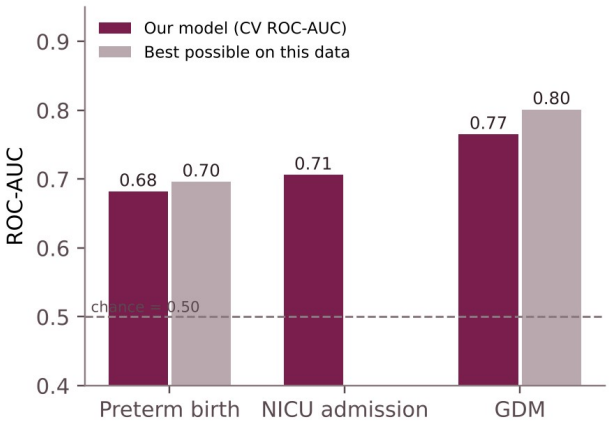


Fig. 4. Model ROC-AUC compared with the best score achievable on this synthetic data.

C. *Important features*

Permutation importance places previous GDM far ahead for GDM (drop in ROC-AUC 0.25), followed by pre-existing diabetes, matching the finding of Artzi *et al.* [6]. For preterm birth the leading features are the number of high-risk visits so far, previous pre-eclampsia, maternal age and BMI; for NICU admission they are previous caesarean section, maternal age, BMI and the drop in haemoglobin.

D. *Decision support in use*

The doctor view (Fig. 5) lists the day's patients with review flags, alerts and tabs for medications, tests and risk. The risk tab shows each probability against its bands with the factors behind it. Applied to the dataset, the safety engine found two allergy conflicts that the medication records marked as passed: a patient with an iron allergy receiving iron, and a patient with an NSAID allergy receiving aspirin (Fig. 6).

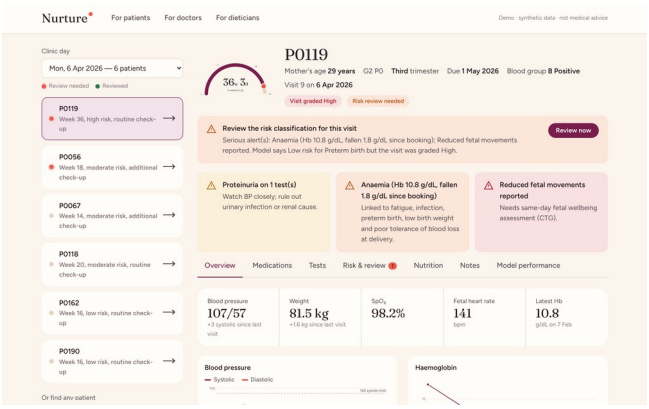


Fig. 5. Doctor view: clinic list with review flags, patient summary, alerts and tabs.

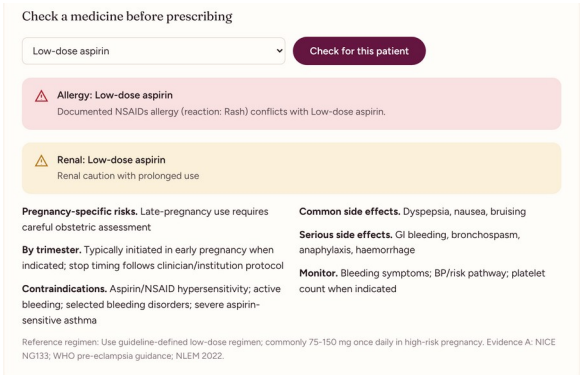


Fig. 6. Prescribing check: an NSAID allergy conflicts with low-dose aspirin.

E. *System responsiveness*

Measured on a single server, start-up takes about 10 seconds; each screen's API call takes 10–130 ms (14 ms when cached) and returns 1–6 KB after compression. Switching between patients in the clinic list takes 70–100 ms because the other patients are pre-loaded. The service uses about 260–330 MB of memory, small enough for free cloud hosting.

VI. Limitations

The data is synthetic, so the results show that the pipeline is correct, not that it is clinically accurate. The cohort is small, with only 14 NICU admissions, and there is no external validation. Preterm birth and NICU admission are largely random in the generator, which caps their achievable accuracy. The rules have not been reviewed by clinicians, explanations by occlusion are approximate, and the demonstration stores data in files without user authentication.

VII. Conclusion and Future Work

Nurture shows that a maternal decision support system can be built so that its predictions are honest: features respect time, validation respects patients, scores are compared with what the data can support, and a doctor confirms every risk classification. Auditing the data pipeline mattered as much as the models, since two generator defects alone would have produced meaningless outcome labels.

Future work will rewrite the outcome generators using effect sizes from published studies, validate on de-identified real records with ethical approval, calibrate the probabilities, move storage to a database with authentication, and offer the patient view in Hindi and Kannada with appointment reminders.

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